

Paper –3: Science of Dental Materials

Departmental Objectives:

After completing this course, the students will be able to:

- Explain general properties [chemical, physical, mechanical and biological] of the materials
- Identify the materials to be used in dentistry
- Describe the composition, properties of the individual material
- Explain the bio-compatibility of the materials
- Discuss the toxic and side effects of the materials and protective measures to be taken
- Describe the manipulation of the materials, its clinical and laboratory uses
- Explain the variables that affect the properties of the final products of the materials
- Discuss the tarnish and corrosion and the preventive measures of the metallic made prostheses and restoratives used in the oral cavity
- Describe the proper handling and caring of the materials
- Select the appropriate material for the clinical use

List of Competencies:

- Gain knowledge on the Selection of the required instruments and its proper use.
- Know the proper use of the instruments.
- Can do to manipulate the dental materials perfectly.
- Can care the materials in the self without loss of its properties.

Hands on training, practical exposure:

1. A fully fledged science of dental laboratory well equipped with the instruments and materials should be established in every institution.
2. The students should be guided by the experienced and expert teachers of the subject concerned
3. A technologist per 15 students should be appointed to the laboratory
4. Practical works should be set based on the total course curriculum and the students must have to perform these practical works as a prerequisite to appear at the final examination.
5. In every term examination, a part of the practical works should be set to perform

Distribution of teaching – learning hours

Lecture	Tutorial	Practical	Demonstration+ card exam	Total teaching hours	Integrated teaching for phase (Common)	Formative exam.		Summative exam.	
						Preparatory leave	Examination time	Preparatory leave	Eax. time
110	40	70	69	289	10 hours	20 days	42 days	30 days	30 days

Teaching – learning method, teaching aid and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture	Demonstration/tutorial		Multimedia/White board/Black board/Hand out	Term i & ii exams., Item exam,

Professional examination:

Marks distribution of assessments

Total marks: 300

Written: 100: [Formative: 10; SAQ: 70 [Group: A-35; Group B-35]; MCQ: 20
[Group: A-10, Group B-10]

SOE: 100

OSPE: 100 [Spotting: 60; Practical: 20; Assignment: Box: 10; practical Khata:10]

Learning Objectives and Course Contents in Science of Dental Materials

Part -1 General classes and properties of dental materials

Learning Objectives	Contents	Teaching Hours
The student will be able to - Classify Dental Materials - Explain the Specification no. of the materials by international association like ADA	1. Overview of Materials for Dental Applications What are Dental Materials, International Standards, US standard for Dental Materials ADA certification	L : 2 Hrs T:1 Hrs
- Describe the different types of bonds - Explain the Bonding forces and Thermal energy	2. Structure of Matter and Principles of Adhesion Change of state, Intra-atomic primary bond, Ionic bond, Covalent bond, metallic bond, inter-atomic secondary bond, hydrogen bonding, vander walls forces, Bonding forces, Thermal energy	L : 3Hrs T:1 Hrs
- Explain the physical properties like Abrasion, abrasion resistance, viscosity, Structural stress relaxation, Creep and flow - Describe the thermal expansion coefficient of material and its dental consideration	3. Physical Properties of Dental Materials What are physical properties, Abrasion and abrasion resistance, viscosity, Structural and stress relaxation, Creep and flow, color and color perception, Thermo-physical properties [Thermal conductivity, thermal diffusivity, coefficient of thermal expansion]	L : 2 Hrs T:1 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to - Describe the mechanical properties of materials like Stress, strains, elasticity, Strength, diametral tensile strength, fracture[toughness, ductility and malleability resilience - Describe the measurement of ductility, hardness - Describe the fatigue and fatigue strength - Explain the importance of proper knowledge on the mechanical properties on the materials using in the construction of restorations/prostheses	4. Mechanical properties What are the mechanical properties, Stress and strains curve[Tensile stress, compressive stress, shear stress, flexural stress], Mechanical properties based on elastic deformation [elastic modulus ,flexibility, resilience, Poisson's ratio] Strength properties [proportional limit, elastic limit, yield strength, Permanent deformation, cold working, diametral tensile strength, flexure strength, fatigue strength, impact strength] Other mechanical properties [toughness, fracture toughness, brittleness, ductility and malleability, measurement of ductility, hardness]	L: 3 hrs T:1 hrs
The student will be able to - Describe the metallic bonds, Solidification of metals - Explain grain refinement and importance of grain size of the alloys	5. Solidification and microstructure of metals Metals, metallic bonds, Alloys, Solidification of metals [Nucleus formation, solidification modes and effects on properties], Grain refinement and grain size	L : 2 Hrs T:1 Hrs
The student will be able to - Classify the alloys - Explain the equilibrium phase diagram - Explain the Solid state reactions of different alloys like Gold copper system, Silver copper system	6. Equilibrium phases in cast alloys Classification of alloys, Solid solution [Solutes and solvents, condition of solid solubility, physical properties of solid solutions], Constitution of equilibrium phase [diagrams, interpretation of the phase diagram, coring, homogenization, dendrite formation in the alloys], Eutectic alloys [Silver copper system], physical properties, Solid state reactions [Gold copper system, Silver copper system] Other binary systems [Gold alloys, palladium alloys]	L : 2 Hrs T:1 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Explain the Applications of resins in dentistry, classification, Requisites for dental polymer ● Describe the fundamental nature of polymers ● Describe Physical properties of polymer ● Describe the Chemistry of polymerization	7. Dental Polymer Applications of resins in dentistry, classification, Requisites for dental resins [Biological compatibility, physical properties, manipulation, Aesthetic properties, economic considerations, chemical stability, fundamental nature of polymers, chain length and molecular weight, chain branching and cross linking, molecular organization], Physical properties of polymers [deformation and recovery, rheometric properties,	L : 3Hrs T: 1 Hrs
The student will be able to ● Describe the adverse effects from dental materials ● Explain the Biological response in the dental environment , ● Describe the Clinical guidelines for selecting biocompatible materials	8. Biocompatibility of dental materials Adverse effects from dental materials [Toxicity, inflammation, Allergy, local and systemic effects of materials] Biological response in the dental environment ,Osseointegration, the oral immune system. Current biocompatibility issues in dentistry [latex, nickel, beryllium, mercury and amalgam, biological effects of resins] Clinical guidelines for selecting biocompatible materials	L : 3Hrs T:1 Hrs

Part II Auxiliary Dental Materials

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Classify impression materials and list the ideal requirements of it. ● Differentiate between elastic and non-elastic impression materials. b. Reversible and irreversible materials ● Describe the setting mechanism, composition of different types of impression materials and their properties ● Identify the most useful materials ● Prepare the custom tray for elastomeric impression material ● Describe the manipulation techniques of different types of impression materials with the effects of water, temp. ● Describe the biocompatibility, disinfection, dimensional stability, compatibility with gypsum, shelf life, effect of mishandling of the impression materials	9. Impression Materials Define Impression, Impression materials, Purpose and requirements, Materials used for making impressions, setting mechanism, mechanical properties, uses of impression materials. Elastomeric impression materials Characteristics, Visco-elastic properties, Elastomeric impression materials: Chemistry and composition, Polysulphide, condensation silicone, Addition silicone, polyether Elastomeric materials: making an impression, preparation of impression materials, impression trays, steps required to make an impression, removal of the impression, preparation of stone cast and die. Elastomeric impression materials: Properties, working and setting times, dimensional stability, reproduction of oral detail, disinfection, rheological properties, elasticity, tear strength, bio-compatibility, shelf life, effect of mishandling, Hydrocolloids: sol to gel transformation, gel strength, dimensional effects, Agar (reversible)hydrocolloids, composition, Manipulation, preparation and conditioning of the agar material, tempering of the material, making the agar impression, accuracy, viscosity of the sol, distortion during gelation Alginate hydrocolloids, composition. Gelation process, controlling of the setting time, manipulation, making the impression, strength, accuracy, laminate technique [alginate-agar method], duplicating materials, modified alginates, biocompatibility, disinfection, dimensional stability, compatibility with gypsum, shelf life, Effect of mishandling, Impression compound: Composition, manipulation, dimensional stability, disinfection Zinc-oxide eugenol Paste: Composition, manipulation, dimensional stability, disinfection, Non-eugenol paste, surgical paste, bite registration pastes	L : 7 Hrs T:2 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to - Describe the uses of gypsum and production. - Discuss the differences between dental plaster and hard plaster - Describe the setting of gypsum products and test of setting expansion, control of setting expansion, accelerators and retarders: practice and theories [Accelerators, retarders] - Discuss ADA classification, the setting mechanism, mixing process and caring	10 . Gypsum Products Uses of gypsum in dentistry, Production of calcium sulphate hemihydrates, setting of gypsum products, Test of setting expansion, control of setting expansion, accelerators and retarders: practice and theories[Accelerators, retarders], Hygroscopic setting expansion, strength, types of gypsum products [impression plaster Type-I, Model plaster-Type II, Dental Stone-Type III, Dental stone High strength type –IV, Dental stone high strength High expansion-Type V, synthetic gypsum], Proportioning, mixing and caring for gypsum products [proportioning, mixing, caring for cast, special gypsum product, caring for gypsum products], Infection control	L : 5 Hrs T: 1 Hrs
The student will be able to - Describe the different types of investment materials and their composition and ideal requirements - Explain the different factors affecting the setting expansion of Gypsum bonded investment materials - Describe the setting reaction of Phosphate bonded investment with its properties, setting reactions - Describe the Ethyl silicate with its Compensation and setting reaction - Describe the Preparation of the master die - Describe the principles of optimal sprue design - Describe the technique involved in Investing procedure Casting procedure - Describe the different steps involved in casting procedure. - Describe the melting and casting machines - Explain the causes of defective castings and its preventive measures	11. Casting Investment and Procedures Definition, Types of investment materials. Gypsum bonded investment materials [composition, setting time,] Normal setting expansion [Effect of different factors on the setting expansion] Thermal contraction, strength, porosity, storage. Phosphate bonded investment: Composition, setting reactions, setting and thermal expansion, working and setting time, miscellaneous properties. Ethyl silicate. Compensation of solidification shrinkage, Preparation of the master die Variables and principles of optimal sprue design, Wax pattern removal, sprue diameter, Sprue position, sprue attachment, Sprue direction, sprue length. Casting ring liner. Investing procedure: vacuum mixing, compensation for shrinkage. Casting procedure Wax elimination and heating, hygroscopic low heat technique, high heat thermal expansion technique. Time allowable for casting, casting machines, torch melting/centrifugal casting machine, Torch melting of noble metal, cleaning the casting, melting of base metal, Technique consideration for phosphate bonded investment, causes of defective castings	L : 6 Hrs T:2 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to - Describe the classification of separating medium - Describe the Properties, application & uses of it in dental field. - Describe the setting mechanism and technique involved in its application	12. Separating medium Chemistry of separating media, Classification, Properties, application & uses in dental field	L : 5 Hrs T: 1 Hrs
The student will be able to - Classify dental waxes - Describe the Types of inlay wax, composition, desirable properties, flow, thermal properties, wax distortion - Explain the manipulation technique involved in different waxes	13. Dental Waxes Classification of dental waxes, Types of inlay wax, composition, desirable properties, flow, thermal properties, wax distortion, manipulation of inlay wax, other dental waxes and their properties and uses	L : 3 Hrs T: 1 Hrs
The student will be able to - Describe the Dental fluxes and its ideal requirements - Explain the Types of dental fluxes and their uses	14. Dental Fluxes Dental fluxes, ideal requirements of dental fluxes, Types of dental fluxes, Uses of dental fluxes	L : 3 Hrs T:1 Hrs
The student will be able to - Describe the classification and necessity of polishing materials - Describe the principles of cutting by the polishing materials - Describe the abrasive instrument design, - Describe the finishing and polishing procedures - Explain the biological hazards of the finishing process - Explain the Dentifrices and its composition	15. Finishing and Polishing Materials Definition, necessity, classification of materials used for these purposes in dentistry. Principles of cutting, grinding, finishing and polishing, abrasion, and erosion, abrasive instrument design, Types of abrasive, finishing and polishing procedures, Biological hazards of the finishing process Dentifrices: composition, abrasiveness, tooth brushes	L : 3Hrs T:1 Hrs

Part-III : Direct Restorative Materials

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Explain the Mechanism of Adhesion ● Describe the Acid Etch technique, ● List the Dentine bonding agents ● Explain the bonding of Glass ionomer restorative, Amalgam bonding ● Describe the Pits and fissures sealers	16. Bonding Mechanism of Adhesion, Acid Etch technique, Dentine bonding agents Measurements of bond strength [micro leakage], Glass ionomer restorative, Amalgam bonding, Pits and fissures sealers	L : 3Hrs T:1 Hrs
The student will be able to ● List the aesthetic restorative materials and their uses ● Describe the Dental composites, its composition and function of components, Resin matrix, Filler particles ● Explain the activator-initiator system ● Explain the chemical activation, light activation, curing lamps, depth of cure and exposure of time ● Describe the Classification of resin based composites	17. Restorative resins [Composite] Aesthetic restorative materials, Uses of restorative materials, Dental composites, composition and function of components, Resin matrix, Filler particles and its benefits, coupling agents, Activator-initiator system, Chemically activated resins, inhibitors, Curing of resin-based composites: chemical activation, light activation, curing lamps, depth of cure and exposure of time, dual-cure resins and extra-oral curing, reduction of residual stresses, incremental building and cavity configuration and delayed curing. Soft-start, ramped curing and delayed curing, precautions for using curing lamps Classification of resin based composites, Traditional composites, Small particle filled composites, clinical consideration of SPF, microfilled composite, flowable composites for posterior restorations[direct and indirect posterior composites], finishing of composites, bio-compatibility of composites, repair of composites. Explain the direct and indirect posterior composites Describe the finishing of composites, bio-compatibility of composites, and repair of composites	L : 5 Hrs T: 2 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Enumerate the different cements used in dentistry ● Describe the composition, properties and uses of different cements ● Describe the manipulation techniques involved in different cements ● Explain the manipulation variables that affects the properties of the cements ● Describe the setting mechanism of some cement. ● Describe the modified forms of glass ionomer cements with their properties, uses	18. Dental Cements and Filling Materials Zinc phosphate cement: Composition, setting, working and setting times, physical properties, retention, biological properties, manipulation. Zinc Polycarboxylate cement: Composition and chemistry, Bonding to tooth structure, film thickness, working and setting times, mechanical properties, solubility, biological preconsideration, manipulation, surface and retention, removal of excess cement Glass ionomer cement composition, chemistry of setting, physical properties, manipulation consideration, surface preparation, preparation of the materials, Placement of the restorative material and removal of the excess material Metal reinforced glass ionomer and resin modified cements: general properties, clinical consideration Hybrid glass ionomer: composition and setting reaction, characteristics of hybrid GIC, fissure sealants applications, liner and base applications, Compomer: composition and chemistry, characteristics of compomer, manipulation of compomers Resin cements: Composition and chemistry, Characteristics, Manipulation, Metallic prosthesis Zinc-oxide eugenol cement: Composition, setting of chemistry, Characteristics of this cement, Temporary ZOE restorations, intermediate ZOE restorations, Temporary ZOE luting cement Calcium hydroxide cement: cavity liners, bases, solubility and disintegration of the cement	L : 6 Hrs T: 2 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Describe dental amalgam and its Classification, composition, ● Explain the metallurgical phase of dental amalgam, ● Explain the dimensional stability, dimensional change of dental amalgam ● Describe the manufacture of alloy powder ● Describe the amalgamation and resulting microstructures ● Explain the manipulation of amalgam ● Explain the effect of moisture contamination, ● Explain the significance of creep on amalgam performance. ● Explain the Influence of manipulative variables on creep. ● Describe the Tarnish and corrosion of amalgam ● Describe the Factors affecting success of amalgam	19. Dental Amalgam Definition of dental amalgam , Classification of Dental amalgam, Alloy composition, Metallurgical phase of dental amalgam, The silver-tin system, dimensional stability, dimensional change, Influence of Ag-Tin phase on amalgam properties, Manufacture of alloy powder, lathe cut powder, homogenization, particle treatment. Atomized powder, particle size, lathe cut powder compared with atomized spherical powder, amalgamation and resulting microstructures, low copper alloys, high copper alloys. Admixed alloys, single composition, Effect of moisture contamination, strength of amalgam hardening rate, condensation, effect of porosity, , effect of trituration, effect of mercury contents, Creep, Significance of creep on amalgam performance. Influence of manipulative variables on creep. Tarnish and corrosion of amalgam, Factors affecting success of amalgam, Clinical significance of dimensional change, Side effects and toxicity of Mercury, Marginal deterioration	L : 6 Hrs T:2 Hrs
The student will be able to ● Describe the types of Gold foils ● Explain the technique involved in direct gold filling & its Properties	20. Direct Gold Filling Metallurgical Principles: Welding, Work hardening Materials; Gold foils sheet, Mat gold, powdered gold, Non-cohesive gold, direct gold alloys Manipulation: degassing, compaction Properties: density, mechanical properties, other properties, Critique of gold foils	L : 2 Hrs T:1 Hrs

Part-IV : Indirect restorative and prosthetic materials

Learning Objectives	Contents	Teaching Hours
The student will be able to - Describe the properties of dental casting alloys and classification - Explain the Karat and fineness, - Describe the Heat treatment techniques involved in High noble and noble metal Alloys - Explain the Casting shrinkage of alloys - Explain the Re-melting of previous casting alloys - Explain the discolouration of porcelain by silver - Explain the thermo-compatibility and incompatibility metal ceramic systems - List the alloys for conventional veneering porcelains and ultra low fusing porcelain	21. Dental Casting and Soldering Alloys Desirable properties of dental casting alloys and classification, Karat and fineness, Alloys for all metal and resin veneered restorations, Heat treatment of High noble and noble metal Alloys, Softening heat treatment of gold casting alloys, Silver and its alloys, Copper and its alloys, Platinum and its alloys, Casting shrinkage, Silver palladium alloys, Silver-nickel and cobalt chromium alloys, titanium and titanium alloys, Hardening heat treatment of gold casting alloys, Re-melting of previous casting alloys, Gold-Palladium- Silver Alloys, Palladium- Gold Alloys, Palladium- Gold-Silver Alloys, Discoloration of porcelain by silver, Thermocompatibility and incompatibility metal ceramic systems, Alloys for conventional veneering porcelains, Alloys for ultra low fusing porcelain	L : 5 Hrs T:2 Hrs
The student will be able to - Describe the Ideal requirements of the alloys of cast partial denture. - Enlisted alloys of cast partial denture - Explain the parts of cast partial denture - Differentiate between Cobalt-chromium alloys and gold alloys as cast partial denture alloys	22. Cast Partial Denture Alloys Ideal requirements, Enlisted alloys of cast partial denture, parts of cast partial denture, Cobalt-chromium alloys, Silver-Palladium Alloys, Type IV gold alloys, Titanium and titanium alloys, Ni-Cr-Be alloys	L : 2 Hrs T:1 Hrs
The student will be able to - Describe the Requirements of alloys for inlays, crowns and bridges - Explain the classification gold alloys - Explain the classification gold alloys, composition and its uses - Describe the heat treatment, mechanical properties of gold alloys - Describe the alloys of silver palladium alloys, nickel-chromium	23. Alloys for Inlays, Crowns and Bridges Requirements, , Gold alloys with 75% noble metal alloys, classification of gold alloys, composition, dental uses, heat treatment, Mechanical properties, silver palladium alloys, nickel- chromium alloys	L : 2 Hrs T:1 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Describe Types of Corrosion and the oral environment ● Explain the Electrolytic cell, galvanic cell ● Explain the factors responsible for corrosion ● Describe the Prevention of galvanic pain	24. Tarnish and Corrosion Definition and Types of Corrosion, The oral environment, Electrochemistry [Electrolytic cell, galvanic cell], Examples of galvanic corrosion [difference of materials, electrolytes, stress, stress corrosion], corrosion and galvanic pain, Prevention of galvanic pain, Electro deposition	L: 2 Hrs T: 1Hrs
The student will be able to ● Describe Types of alloys for PFM prostheses and their Requirements ● Explain the Cooling technique of metal –Ceramic Prosthesis ● Explain Creep or sag resistance ● Explain the mechanisms involved in Bonding of Porcelain to metal ● Explain Benefits and drawbacks of metal ceramics	25. Porcelain Fused to Metal Introduction, Types of alloys for PFM prostheses, Requirement, Aesthetic potential of Metal-Ceramic Crowns versus All ceramic Crowns, Overglazing and shading of Ceramics, Cooling of metal –Ceramic Prosthesis, Creep or sag resistance, Coping of metal ceramic prostheses, Bonding of Porcelain to metal, Benefits and drawbacks of metal ceramics The alloys designated for PFM Prostheses: Gold, alloy, cobalt chromium alloy, Silver palladium alloys Nickel Chromium alloys, Choice of Ceramic	L : 3 Hrs T:1 Hrs
The student will be able to ● Explain wrought alloys ● Annealing of cold worked metals, ● Describe sensitization and prevention of S.S. wires, , ● Soldering and welding of stainless steel, corrosion resistance properties of austenitic stainless steel ● Describe the orthodontic wire alloys composition, its super-elasticity and shape memory, ● Describe mechanical properties of beta-titanium wires	26. Wrought alloys Introduction of wrought alloys and orthodontic wires, point defect, dislocations, Effects of annealing cold worked metals, Carbon steels, stainless steel, corrosion resistance properties of austenitic [stainless steel, sensitization, stabilization, general causes of corrosion, mechanical properties, recovery heat treatment, braided and twisted wires, soldering and welding of stainless steel wire, nickel titanium alloys[orthodontic wire alloys composition, super-elasticity and shape memory, nickel titanium endodontic instruments], beta-titanium alloys, titanium alloys, mechanical properties of beta-titanium wires, welding, corrosion resistance, noble metal wrought wire	L : 2 Hrs T: 1Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to ● Describe the Definition, Classes of Dental Ceramics and its Composition ● Explain the techniques involved in Porcelain Condensation ● Describe the different Methods involving in strengthening of Ceramics ● Explain the Abrasiveness of Dental Ceramics, Wear of ceramics and Guidelines for minimizing excessive Wear of enamel by Dental Ceramic ● Describe Clinical Performance of Ceramic Prosthesis ● Describe the Criteria for Selection and Use of Dental Ceramics	27. Dental Ceramics Definition, Classes of Dental Ceramics, Ceramic Processing Methods, Composition of Dental Porcelains, Glass modifiers, Feldspathic porcelain, Other additives, Porcelain Condensation, Sintering of Porcelain, Ceramic Prostheses [Aluminous Porcelain Crowns, Castable and Machinable Glass-Ceramics, CAD-CAM Ceramics, Methods of strengthening Ceramics[Minimize the Effect of stress Raisers, Develop Residual Compressive Stresses, Minimize the Number of firing Cycles, Ion Exchange, Thermal tempering, Dispersion Strengthening], Abrasiveness of Dental Ceramics, Wear of ceramics Compared with other Materials, Wear of Enamel by Ceramic Products and other Restorative Materials, Reducing Abrasiveness of Ceramics by Polishing and Glazing, Guidelines for minimizing Excessive Wear of enamel by Dental Ceramic s, Clinical Performance of Ceramic Prosthesis, Porcelain Denture teeth, Factors affecting the color of Ceramics, Criteria for Selection and Use of Dental Ceramics	L : 6 Hrs T:2 Hrs
The student will be able to ● Define die, counter die, Swaging ● Describe the construction of S.S, denture base ● Explain the differences between metallic denture base and non metallic denture base	28. Alloys for Die and Counter Die Definition of die, counter die, Swaging, construction of S.S, denture base, difference between metallic denture base and non metallic denture base	L : 3 Hrs T:1Hrs
The student will be able to ● Classify and define solders, soldering, welding. ● Explain Ideal requirements of solder and its different types and composition ● Describe the heating sources, zones of a flame ● Describe the techniques involved in the different joining processes ● Explain the successful soldering, and causes of failure ● Explain the causes of defective soldering , antifix used in soldering processes, technique of S.S. wire soldering	29. Soldering, Brazing, and Welding Definition[solders, soldering, welding, brazing], Ideal requirements of solder, Types of solders, Composition, properties of solder, heating sources, zones of a flame, types of soldering process[techniques], requirement of successful soldering, Principles of soldering, causes of failure, antifix , pitted soldering, S.S. wire soldering, Buckling, welding, types of welding, welding technique	L : 3 Hrs T:1 Hrs

Learning Objectives	Contents	Teaching Hours
The student will be able to • Classify dental implants • Describe the implant design, • Explain the properties of implant and its attachment mechanisms • Describe the different implant components and success and failure of dental implants • Describe different types of implant materials • Describe the selection criteria of implant materials • Describe its biocompatibility and biomechanics	30. Dental Implants Classification of dental implants, implant design, implant properties, indications and contraindications, attachment mechanism, implant components, clinical success of dental implants, implant materials, metallic implant, ceramic and ceramic coated implants, selection of implant materials, biocompatibility of implants, biomechanics	L : 3 Hrs T:1 Hrs
The student will be able to • Classify denture base Materials • Enumerate the Requirements of polymeric denture base materials • Summarize the fundamental nature of polymer • Describe the components of Heat activated denture base resins and its storage • Describe the steps involved in the construction of a denture • Explain the causes of development of different defects during curing and their preventive measures • Describe the Chemically activated resins and Light Activated resin and technique of manipulation • Describe the technique involved in Repair Resins, Relining Resins Denture Bases, Rebasing Resin Dentures • Describe the infection control procedures, hygienist's role in maintenance of denture base and allergic reactions	31. Denture Base Resins Classification of denture base materials, polymeric denture base materials.: Requirements of polymeric denture base materials. Fundamental nature of polymer, Heat activated denture base resins [composition, storage], compression moulding technique [preparation of the mould, selection and application of the separating medium], Polymer-monomer interaction [dough time, working time, packing, injection moulding technique, polymerization procedure, temperature rise, internal porosity, polymerization cycle] Chemically activated resins, [technical consideration, processing considerations], Light Activated Denture Base Resins, Physical Properties of Denture Base Resins, [Polymerization shrinkage, Porosity, Water absorption, Solubility, Processes stresses, Crazeing, Strength, Creep, Miscellaneous properties], Repair Resins, Relining Resins Denture Bases, Rebasing Resin Dentures, Short term and long term soft denture liners, resin impression tray materials, Denture cleansers, infection control procedures, Allergic reactions, Toxicology	L : 5 Hrs T:2 Hrs
The student will be able to • Describe the importance, properties of the important base metals used in dentistry	32. Base Metal and its Importance Base metal like Zn, Ni, Cr, Pb Its importance in dentistry, sources, properties and dental use of base metals	L : 2 Hrs T:1 Hrs

Practical phase on Science of Dental Materials

Related equipments:

Purpose	Related instruments
1. Manipulation of different types of impression materials	Rubber bowl, Plaster spatula, impression trays of different types [Plastic; disposable, stainless steel perforated, plain], Mixing gun [elastomeric impression materials], Sprit lamp, Bunsen burner, Wax knife, Wax carver, Copper band, Glass slab or oil impervious paper, Steel spatula [Broad blade, Stiff but flexible]. For Agar impression Material: Conditioning unit, Rim lock trays
2. Manipulation of gypsum products	Plaster spatula, Rubber bowl, Glass slab, Mechanical mixer
3. Application of separating media	Camel hair brush
4. Preparation of wax pattern	Wax knife, wax carver, plaster knife, sprit lamp, Bunsen burner
5. Manipulation of acrylic resin and construction of denture base [for practice on model]	A narrow porcelain container, Dental flask, Press
6. Manipulation of cement [different types]	Cement spatula, Glass slab, mixing pad, Ball ended instrument [for calcium hydroxide cement]
7. Manipulation of composite resin	[for practice on model]: Paper pad, plastic spatula, Light curing unit, S.S. spatula
8. Manipulation of amalgam filling	Pestle and mortar [hand mixing], Amalgamator [mechanical mixing], Amalgam gun, Condenser, Amalgam carver, Polisher
9. Casting procedure	Casting ring, ring liner, crucible former, crucible, vibrator, Preheating furnace, casting machine [centrifugal], Gas burner, Gas cylinder [titanium and oxygen]
10. Manipulation of dental porcelain	S.S. spatula, camel hair brush [special brush to condense porcelain slurry mix is available], special mixing pot [ceramic], Preheating furnace, Porcelain furnace,
11. For polishing purpose	Hanging mortar, a grinding mortar, Turbine hand piece, slow speed hand piece, Diamond bur of different sizes, polishing rubber cup, polishing agents
12. Auxiliary works [to prepare practicing models]	Mould of Edentulous, Dentulous and partial Denture

List of Manipulative works

Sl. No.	Name of the materials	Learning method	Teaching aids and expected hours
1	Manipulation of compound impression material [Sheet Variety]. Taking impression with Compound impression [materials & Phantom head or mouth].	Demonstration on composition, uses, properties and required instruments in SDM lab.	Aids: Multimedia Expected hours: 2
2	Manipulation of Low fusing compo. Seal the Boarder of the special tray.	Demonstration on Identification of low fusing compo, its properties uses and manipulation process and on the required instruments.	Aids: Multimedia Expected hours: 2
3	Copper band impression technique by low fusing compo.	Demonstration on Identification of different types of materials to be used as copperband impression materials and their other uses and manipulation process. Demonstration on copperband and the required instruments.	Aids: Multimedia Expected hours: 2
4	Manipulation of Zinc oxide eugenol paste and Taking impression of on an edentulous model.	Demonstration on Identification of Zinc oxide eugenol pastes, their uses, properties composition and manipulation process. Identification of the required instruments.	Aids: Multimedia Expected hours: 2
5	Manipulation of alginante.& Taking impression of an edentulous model.	Demonstration and Identification of alginate powder, its properties, uses and manipulation process. Demonstration on different types of impression trays and how to take an impression with alginate and manipulation process of alginate.	Aids: Multimedia Expected hours: 3
6	Preparation of custom tray with self cured acrylic and take an impression of a model.	Demonstration on custom tray with its purposes. Demonstration on a model to show technique of manipulation of a custom tray by self cured acrylic resins.	Aids: Multimedia Expected hours: 2

Sl. No.	Name of the materials	Learning method	Teaching aids and expected hours
7	Elastomeric impression material: Two stage putty wash technique.	Demonstration on an elastomeric impression materials with its composition, uses, properties and technique to load a custom made tray by elastomeric impression material to take an impression.	Aids: Multimedia Expected hours: 3
8	Manipulation of gutta percha and loading into a copper band tray to take an impression of a prepared tooth.	Demonstration on different forms of gutta percha with its uses, properties and composition. Demonstration on the technique to load copper band tray with the manipulated stick gutta percha to take an impression of a single prepared tooth.	Aids: Multimedia Expected hours: 2
9	Manipulation of plaster of paris :Hand mixing and mechanical mixing.a). Cast an impression taken either by alginate or compo.	Identification of different types of gypsum products, their properties, composition and technique to manipulate plaster of paris and how to cast an impression to make a model .	Aids: Multimedia Expected hours: 3
10	b). Test of setting time by setting time tester.	Demonstration on base plate waxes and a readymade special tray/ temporary base. Discussion on purposes of a special tray/ temporary base and uses, composition, properties of base plate wax with its manipulation technique.	Aids: Multimedia Expected hours: 3
11	Manipulation of a Base plate wax to make a special tray/temporary base	Demonstration on modeling wax with its uses, composition, properties. Show a ready made wax pattern and wax rim made on special tray. Discuss the technique to manipulate a modeling wax piece to make a wax pattern on edentulous model.	Aids: Multimedia Expected hours: 3
12	Manipulation of modeling wax to make a wax pattern.	Demonstration on different forms of inlay casting waxes and ready made pattern wax for casting purpose.	Aids: Multimedia Expected hours: 3

Sl. No.	Name of the materials	Learning method	Teaching aids and expected hours
13	Manipulation of sticky wax [Joining of the fractured parts by softened sticky wax].	Demonstration on a piece of sticky wax. Demonstration on a joined fractured part with sticky wax. Discuss on the uses and properties of sticky wax.	Aids: Multimedia Expected hours: 3
14	Application of cold mould seal on a plaster model.	Demonstration on the required instruments. Demonstration on cold mould seal and some other separating media like vaseline. Discuss the composition, setting reaction and purpose of the use of separating media like cold mould seal.	Aids: Multimedia Expected hours: 3
15	Manipulation of acrylic resin [dough adaptation technique].	Demonstration on acrylic resins on its uses, properties and manipulation technique. Show different stages as recognized during its manipulation.	Aids: Multimedia Expected hours: 3
16	Technique to construct a denture base by acrylic resin.	Demonstration on the different instruments necessary to construct a denture base. Demonstrate how a defect free denture base can be constructed. Demonstration on the different types of porosities with their preventive measures.	Aids: Multimedia Expected hours: 2
17	Technique to repair fractured denture/model.	Practically demonstrate on the technique to repair the two fractured parts of a model or denture for subsequent repairing procedures. Show why sticky wax is good for repairing.	Aids: Multimedia Expected hours: 2
18	Application of varnish.	Identification of the instruments required to application of varnish. Demonstration on varnish emphasizing on its importance, composition and technique to varnish the dentinal walls of a prepared cavity.	Aids: Multimedia Expected hours: 1

Sl. No.	Name of the materials	Learning method	Teaching aids and expected hours
19	Manipulation of Zinc oxide eugenol cement.	Demonstration on the required instruments and materials required to prepare a temporary filling or temporary luting cement.	Aids: Multimedia Expected hours: 3
20	Manipulation of calcium hydroxide cement.	Demonstration on required instruments. Demonstration on the two tubes of calcium hydroxide and also on the powder of the cements. Discuss on the uses, composition, properties and technique to manipulate the cement.	Aids: Multimedia Expected hours: 3
21	Manipulation of zinc oxyphosphate cement.	Demonstration on the required instruments. Discuss on the uses, composition and properties of this cement. Demonstrate the manipulation technique on the cement.	Aids: Multimedia Expected hours: 2
22	Manipulation of Glass ionomer cement.	Demonstration on the required instruments. Discuss on the uses, composition and properties of this cement. Demonstrate the manipulation technique on the cement.	Aids: Multimedia Expected hours: 2
23	Manipulation of polycarboxylate cement.	Demonstration on the required instruments. Discuss on the uses, composition and properties of this cement. Demonstrate the manipulation technique of the cement.	Aids: Multimedia Expected hours: 2
24	Manipulation of light cured composite resin. Experiment: a) Effect of curing time on the depth of composite layer.	Demonstration on the composite. Identification on the required instruments to restore a cavity by light cured resin.	Aids: Multimedia Expected hours: 2

Sl. No.	Name of the materials	Learning method	Teaching aids and expected hours
25	Manipulation of self cured composite resin.	Demonstration on the resin and the required instruments.	Aids: Multimedia Expected hours: 2
26	Manipulation of amalgam alloy: Hand mixig and mechanical mixing.	Demonstration on the different forms of amalgam alloy available. Demonstration on the required instruments and techniques [mechanical and hand mixing]. Demonstration on Hg about its toxicity and preventive measures.	Aids: Multimedia Expected hours: 2
27	Introduction to cast partial denture frame work and denture.	Discuss on this frame work. Give a brief note on the differen alloys used for this purpose. Demonstrate the different parts of the PD denture as identified in this framework.	Aids: Multimedia Expected hours: 2
28	Casting procedure.	Demonstration on the different instrument required to carry out casting and its different uses. Demonstrate on the investment materials. Give the technique to invest the wax pattern for burn out process. Burnout of the wax pattern and carryout casting process by practicing metal.	Aids: Multimedia Expected hours: 2
29	Free hand soldering.	Demonstration on the free hand soldering process and the soldering machine. Technique to solder S.S. wire.	Aids: Multimedia Expected hours: 1
30	Spot welding.	Demonstration on the spot welding. Instruments requiredfor this purpose.	Aids: Multimedia Expected hours: 2

Assignment :(Minimum no. of Contents of Practical Box)

1. One Pair of Models [Edentulous -1, Dentulous- Upper/Lower]
2. Temporary Base by base plate wax (Upper/Lower)
3. Wax Pattern with Modeling Wax (Upper/Lower)
4. Special tray Made by Self cured acrylic resin (upper/lower)
5. Joining of broken Model by sticky wax
6. Fill up a Class-I cavity on model by Amalgam filling